Inorganic Chemistry CUME
March-31-2017

PART-I. BASICS: *(10 points for each)*

1. In the van der Waals forces, which is the strongest and which is the weakest?

2. Find one anion and one cation that both are isoelectronic to SiH_4.

3. Find one typical III-V semiconductor, and one typical II-VI semiconductor.

4. From Sr to Ag in the Periodic Table, which element has the highest melting point and which is the least reactive?

5. If an aqueous solution's pH = 15, find the $[OH^-]$ and pOH.

Part-II. Questions in Advanced Inorganic Chemistry: *(10 points for each)*

1) Allotropes are different crystalline or molecular forms of the same element. Discuss chemical bonding for each of the carbon-allotropes, and explain why they are colored (or colorless).

2) Explain which carbon-allotrope should be paramagnetic, and which should be diamagnetic.

3) Compare graphite with boron nitride, discuss their similarity/difference in chemical bonding, structure, and chemical property.

4) Compare graphite with boron nitride, discuss their similarity/difference in electrical conductivity, heat conductivity, and color.

5) Explain why boron nitride can (or cannot) form a buckyball structure, and a nanoscale tubular structure.